

Differential Equations for Master Integrals

Lecture notes on the analytic evaluation of the double-real phase-space integrals in
 $pp \rightarrow h + X$ at NNLO

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Abstract

Integration-by-parts reduction expresses all phase-space integrals of the double-real channel through a finite basis of master integrals. These notes develop, from first principles, the method by which the masters are evaluated: the derivation of first-order differential equations in the kinematic invariants; the organization of the equations into triangular systems, and the equivalence relation under which the many systems collapse to a short list of genuinely distinct ones; the canonical form of the equations, in which the Laurent expansion in the dimensional regulator integrates order by order into iterated integrals; the treatment of square roots of kinematic polynomials by rational changes of variables and, where none exists, by the Galois theory of the roots; the determination of boundary conditions from the analytic and geometric properties of the phase space; and the recovery of the distributional behaviour of the integrals at the edges of phase space. Each concept is defined before it is used, each mechanism is derived, and each is illustrated by an example small enough to verify by hand.

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1 Introduction

1.1 Phase-space integrals as cut loop integrals

The physical quantity behind these notes is the NNLO hard function for single-inclusive hadron production, $pp \rightarrow h + X$, in collinear factorization. In its double-real channel two unresolved massless partons are radiated and integrated over, so the basic objects are phase-space integrals of squared tree amplitudes. It is convenient to convert them into loop integrals at the outset. By the Sokhotski–Plemelj identity,

$$\delta^{(+)}(p^2) = \frac{1}{2\pi i} \left[\frac{1}{p^2 - i0} - \frac{1}{p^2 + i0} \right] \quad \text{on positive-energy configurations,} \quad (1)$$

each on-shell delta function is the discontinuity of a propagator; the positive-energy condition $\theta(p^0)$ is supplied by the cut itself rather than by the discontinuity. Every phase-space integral thereby becomes a two-loop integral in which certain lines are *cut*. The value of the rewriting — this is the device known as reverse unitarity — is that cut integrals satisfy the same integration-by-parts identities as ordinary loop integrals, with one additional rule: an integral in which a cut propagator has been cancelled vanishes. All of loop technology thus becomes available to phase-space integration.

After projection onto the twist-two correlators, the integrals depend on the two dimensionless Mandelstam ratios

$$v = -\frac{t}{s}, \quad w = -\frac{u}{s}, \quad (2)$$

whose physical region is the open triangle

$$0 < v, \quad 0 < w, \quad v + w < 1. \quad (3)$$

The Born configuration sits on the line $v + w = 1$, so $1 - v - w \rightarrow 0$ is the soft limit, while $v \rightarrow 0$ and $w \rightarrow 0$ are the two collinear limits; all three edges of the triangle carry physics, and they will reappear throughout, from the singularities of the differential equations to the boundary conditions and the endpoint distributions. Everything is dimensionally regularized, $d = 4 - 2\varepsilon$.

1.2 What is to be computed

Reduction by integration-by-parts identities leaves a basis of 347 *master integrals* $I_m(v, w; \varepsilon)$. The reduction is exact linear algebra over the field $\mathbb{Q}(v, w, \varepsilon)$, but it says nothing about the values of the basis elements. The task is their *analytic evaluation*: for each master, a Laurent expansion

$$I(v, w; \varepsilon) = \sum_{k \geq k_0} \varepsilon^k I^{(k)}(v, w) \quad (4)$$

whose coefficients are explicit transcendental functions — in the end, $\mathbb{Q}$ -linear combinations of iterated integrals of logarithmic differential forms — together with exactly known constants.

The method rests on one mathematical fact: *the masters satisfy first-order linear differential equations in v and w* . Everything else is the systematic exploitation of that fact, and the chapters build on one another accordingly. Section 2 derives the differential equations and their two structural properties, triangularity and integrability. Section 3 introduces the equivalence relation under which the many systems of the problem reduce to a short list of genuinely distinct ones. Section 4 presents the canonical form of the equations, shows that it reduces integration to quadratures, and reads off the local behaviour of the solutions at the singular loci. Section 5 constructs the transformation into canonical form. Section 6 treats the square roots that the kinematics forces on some of the systems. Section 7 assembles the general solution, and Sections 8 and 9 determine the boundary data and the behaviour at the edges of phase space. A closing section reflects on the respective roles of search and proof, and a glossary collects the vocabulary.

2 Differential equations for master integrals

2.1 Masters and integration by parts

Fix a set of inverse propagators $D_1, \dots, D_N$ — including the cut ones — large enough that every scalar product of loop and external momenta is a linear combination of the D_i and external invariants. Such a set, an *integral family*, defines the integrals

$$I_{a_1 \dots a_N}(v, w; \varepsilon) = \int \prod_{\ell=1}^2 \frac{d^d k_\ell}{i\pi^{d/2}} \frac{1}{D_1^{a_1} \dots D_N^{a_N}}, \quad a_i \in \mathbb{Z}, \quad (5)$$

where for a cut line the factor $1/D_i$ stands for the corresponding delta function, held at unit power throughout; an integral in which a cut power has been lowered to zero vanishes by the rule of the introduction. The *sector* of an integral is the subset of propagators present with positive power; deleting propagators leads to *subsectors*, and the sectors of a family form a lattice under this deletion. Integration-by-parts identities,

$$0 = \int \prod_{\ell} d^d k_\ell \frac{\partial}{\partial k_j^\mu} \left[\frac{q^\mu}{D_1^{a_1} \dots D_N^{a_N}} \right], \quad q \in \{k_1, k_2, p_i\}, \quad (6)$$

relate integrals with shifted indices; solving them expresses every integral of the family through a finite basis of master integrals, distributed over the sectors of the family.

2.2 Derivation of the differential system

Two observations combine into differential equations for the masters. A derivative with respect to an invariant can be realized as a combination of derivatives with respect to external momenta acting under the integral sign; these raise and lower propagator powers, producing integrals *of the same family*. Integration by parts then maps those integrals back to the masters. Collecting the masters of a family and its subsectors into a vector $\vec{I}$, one obtains a closed linear system

$$\partial_v \vec{I} = A_v(v, w; \varepsilon) \vec{I}, \quad \partial_w \vec{I} = A_w(v, w; \varepsilon) \vec{I}, \quad (7)$$

with matrices of rational functions. It is convenient to combine the pair into the matrix-valued one-form $A = A_v dv + A_w dw$, the *connection*, and to write $d\vec{I} = A \vec{I}$.

Example 2.1 (The one-mass bubble). *The mechanism in miniature. Let*

$$B(s; \varepsilon) = \int \frac{d^d k}{i\pi^{d/2}} \frac{1}{k^2 (k+p)^2}, \quad s = p^2, \quad (8)$$

a one-scale family with $D_1 = k^2$, $D_2 = (k+p)^2$ and a single nontrivial master, B itself. Differentiate with respect to s using $\partial_s = \frac{1}{2s} p^\mu \partial_{p^\mu}$:

$$\frac{\partial B}{\partial s} = -\frac{1}{2s} \int \frac{d^d k}{i\pi^{d/2}} \frac{2p \cdot (k+p)}{k^2 [(k+p)^2]^2}. \quad (9)$$

The numerator is a combination of inverse propagators, $2p \cdot (k+p) = D_2 - D_1 + s$, so the right-hand side involves only integrals of the family: $I_{1,1} = B$, the scaleless $I_{0,2} = 0$, and $I_{1,2}$. The integration-by-parts identity generated by $\partial_{k^\mu}(k^\mu/D_1 D_2)$ reads $0 = (d-3)I_{1,1} + sI_{1,2}$, that is,

$$I_{1,2} = -\frac{d-3}{s} I_{1,1}, \quad (10)$$

and therefore

$$\frac{\partial B}{\partial s} = -\frac{1}{2s} \left[B - (d-3)B \right] = -\frac{4-d}{2s} B = -\frac{\varepsilon}{s} B. \quad (11)$$

The equation determines the entire s -dependence,

$$B(s; \varepsilon) = c(\varepsilon) (-s)^{-\varepsilon}, \quad (12)$$

with the branch fixed by the causal prescription $s \rightarrow s + i0$ — consistent with dimensional analysis, since B carries mass dimension $d-4 = -2\varepsilon$. A single direct evaluation fixes the remaining constant,

$$c(\varepsilon) = \frac{\Gamma(1+\varepsilon) \Gamma(1-\varepsilon)^2}{\varepsilon (1-2\varepsilon) \Gamma(1-2\varepsilon)}. \quad (13)$$

A differential equation that fixes the functional dependence, and one boundary constant fixed by an elementary integral: the entire subject in one line.

2.3 Triangular structure

Order the sectors of a family by their number of propagators. Differentiation and reduction never lead from a sector to one with more propagators; they lead sideways and down the sector lattice. With the master vector ordered accordingly, the connection is *block lower triangular*,

$$A = \begin{pmatrix} a_{11} & & & \\ a_{21} & a_{22} & & \\ \vdots & & \ddots & \\ a_{b1} & a_{b2} & \cdots & a_{bb} \end{pmatrix}, \quad (14)$$

where each diagonal entry a_{ss} — a *block*, in the terminology fixed in the next section — couples the masters of one group of sectors among themselves, and the entries below the diagonal express how lower sectors source higher ones. The diagonal blocks carry the irreducible analytic content of the system; the entries below the diagonal can be dealt with afterwards — the observation on which Section 5 rests.

2.4 Integrability

Equality of mixed partial derivatives, $\partial_v \partial_w \vec{I} = \partial_w \partial_v \vec{I}$, forces the compatibility condition

$$\partial_v A_w - \partial_w A_v - [A_v, A_w] = 0, \quad \text{equivalently} \quad dA = A \wedge A, \quad (15)$$

the statement that the connection is *flat*. (To check the sign: $\partial_v \partial_w \vec{I} = \partial_v (A_w \vec{I}) = (\partial_v A_w) \vec{I} + A_w \partial_v \vec{I}$, and antisymmetrizing in v, w gives $\partial_v A_w - \partial_w A_v + [A_w, A_v] = 0$.) Flatness plays three roles. It is a stringent consistency condition on any derived or transformed system, since it interlocks all matrix entries. It guarantees that solutions continued along a path depend only on the endpoint and on the homotopy class of the path in the complement of the singular loci, which underlies Section 7. And it is stable under change of basis, so once established for a system it holds for every equivalent form of that system.

3 Families, blocks, and equivalence

The double-real channel does not present one large differential system but many small ones, and many of the small ones turn out to be the same system in disguise. Two equivalence relations — one among integral families, one among diagonal blocks — identify the disguises, and together they set the true size of the analytic problem.

3.1 Equivalence of integral families

Families are generated mechanically, one propagator set per diagram, and diagrams proliferate: the channel begins with 1898 of them. Most are redundant, because different routings of the loop momenta describe the same set of denominators.

Definition 3.1 (Equivalent families; representative families). *Two integral families are equivalent if a unimodular integer redefinition of the loop momenta, composed with shifts by external momenta, maps one propagator set bijectively onto the other and maps cut lines to cut lines. Such a substitution leaves the integration measure invariant, so equivalent families have identical integrals up to a relabeling of indices. A chosen representative of each equivalence class is called a representative family.*

This is where the “shifting of momenta” enters the subject. Equivalence is decided without searching over shift matrices: the Symanzik polynomials of a family, brought to a normal ordering of the Feynman parameters by Pak’s algorithm (with cut lines distinguished by auxiliary labels), identify a family up to relabeling of the parameters. Coincidence of the normal orderings is therefore a reliable *candidate* for a momentum shift; every coincidence is then confirmed by constructing the shift explicitly and applying it, which is what turns the identification into a proof. The 1898 raw families collapse by better than a factor of four, and only a fraction of the survivors supports master integrals at all: ninety-one families, in the end, carry the differential systems of the problem.

Differentiation is family-local — the derivative of a master of a family reduces to masters of that family and its subsectors, never to unrelated ones — so the differential-equation problem separates into 91 independent systems, of dimensions between 2 and 47, accounting for all 347 masters. (The dimensions sum to more than 347: each system contains, besides its own masters, the subsector masters it couples to, and those are counted once, in the family that owns them.) The boundaries of the systems were thus fixed already at the reduction stage; what must be constructed anew is, for each family, the connection (A_v, A_w) , together with the proof of its flatness.

3.2 Blocks and their equivalence

Definition 3.2 (Block). *A block of a family’s system is a strongly connected component of the coupling structure of its connection: a minimal set of masters each of which appears in the derivative of another. With the masters in sector order, the blocks are the diagonal entries a_{ss} of Eq. (14). The 91 family systems contain 1119 blocks.*

Definition 3.3 (Equivalence of blocks; classes). *Two blocks, in the same family or in different ones, are equivalent if their connections coincide as rational matrices after a permutation of the basis, optionally composed with the involution $v \leftrightarrow w$. An equivalence class of blocks is called a class; each class is represented by a single block, whose canonical form is constructed once and serves for all members. The 1119 blocks fall into 173 classes.*

The roughly tenfold collapse has a physical origin. A block is the differential system of the masters of one subtopology, and the same subtopology occurs as a subdiagram of many different parent diagrams: the identification of *complete* families by momentum shifts leaves untouched the far more frequent coincidence of their *subsectors*. The involution $v \leftrightarrow w$ is the crossing symmetry $t \leftrightarrow u$ of the underlying process, which maps cut diagrams into cut diagrams. Equivalence is never assumed on such grounds, however: a candidate identification is accepted only when the permutation and swap reproduce the representative’s matrices identically.

The consequence for the whole method is that the analysis of Sections 5 and 6 is carried out once per class rather than once per block — a reduction of roughly tenfold — after which each family is reconstructed from the canonical forms of its classes. The dictionary — which rows of a family realize which class, and under which permutation and swap — is part of the statement being proved, and is checked again when the family is put back together.

4 The canonical form

4.1 The ε -form and its immediate consequences

Definition 4.1 (Canonical form; letters, alphabet, residues). *A differential system is in canonical form (equivalently, in ε -form; both names are standard) if its connection is*

$$d\vec{F} = \varepsilon \sum_a R_a d \log \varphi_a(v, w) \vec{F}, \quad (16)$$

where the residue matrices R_a are constant — independent of kinematics and regulator alike — and the letters φ_a are functions of the kinematics alone. The collection $\{\varphi_a\}$ is the alphabet of the system. In the simplest and most common case the letters are polynomials; Section 6 enlarges the notion to algebraic functions.

One family of the double-real channel, a system of dimension fourteen, carries the six-letter alphabet

$$\{v, w, 1-v, 1-w, v+w, 1-v-w\}, \quad (17)$$

and every entry of its canonical connection is ε times a rational combination of logarithmic differentials of these six polynomials. Richer systems have richer alphabets — among the two-root families one finds letters quadratic in the variables — but always of the same form.

Three consequences follow at once, and they are linked. The connection has simple poles exactly on the loci $\varphi_a = 0$ and nowhere else — the system is *Fuchsian*, and for generic ε each letter with non-vanishing residue is a genuine branch locus of the solutions, so the alphabet is a complete list of the places where anything can happen. Because each $d \log \varphi_a$ is a closed form, the flatness condition (15) collapses to commutator relations $\sum_{a < b} [R_a, R_b] d \log \varphi_a \wedge d \log \varphi_b = 0$ among constant matrices, evaluated once and for all. And, as the next subsection shows, the coefficient of ε^k in the solution is a k -fold iterated integral, so the expansion is homogeneous in *transcendental weight* — the grading on functions and numbers under which a logarithm carries weight one, a k -fold iterated integral of logarithmic forms weight k , and the constants ζ_k and π^k weight k , with rational numbers at weight zero. In the canonical basis this grading is manifest, order by order in ε .

4.2 Order-by-order integration

Why one wants the form (16) is best said first. In a generic basis the regulator is entangled with the kinematics in every entry of A , and the coefficients of the ε -expansion obey coupled differential equations with no distinguished structure. In the canonical basis the regulator counts the number of integrations and does nothing else: the equations integrate themselves.

To see this, write $\hat{A} = \sum_a R_a d \log \varphi_a$, so that $d\vec{F} = \varepsilon \hat{A} \vec{F}$ with $\hat{A}$ independent of ε . Substituting $\vec{F} = \sum_k \varepsilon^k \vec{F}^{(k)}$ and comparing powers of ε :

$$d\vec{F}^{(0)} = 0, \quad d\vec{F}^{(k)} = \hat{A} \vec{F}^{(k-1)} \quad (k \geq 1). \quad (18)$$

The system has disentangled into a recursion of quadratures: the lowest order is a constant vector, and each subsequent order is an integral of known functions. Along a path γ from a base point p_0 ,

$$\vec{F}^{(k)}(p) = \vec{F}^{(k)}(p_0) + \int_{\gamma} \hat{A} \vec{F}^{(k-1)}, \quad (19)$$

and unwinding the recursion expresses the full solution as the path-ordered exponential

$$\vec{F}(p) = \mathbb{P} \exp \left(\varepsilon \int_{\gamma} \hat{A} \right) \vec{F}(p_0) = \sum_{k \geq 0} \varepsilon^k \sum_{a_1, \dots, a_k} R_{a_k} \cdots R_{a_1} \int_{0 \leq t_1 \leq \dots \leq t_k \leq 1} \prod_{i=1}^k d \log \varphi_{a_i}(\gamma(t_i)) \vec{F}(p_0). \quad (20)$$

The k -fold integrals are *Chen iterated integrals* over the alphabet; flatness makes them independent of the choice of path within its homotopy class. The analytic problem is thereby solved in principle, and what remains is to express the iterated integrals in a standard basis of functions (Section 7) and to determine the boundary vector $\vec{F}(p_0)$ (Section 8).

One apparent paradox is worth resolving immediately. The masters themselves have poles in ε — their Laurent expansions begin at negative orders — while the canonical recursion (18) starts from a *constant* at order ε^0 . There is no contradiction: the poles reside in the transformation between the two bases. The matrix T relating the derived masters to the canonical ones carries inverse powers of ε from the normalization of the basis, so a finite canonical order feeds several Laurent orders of a master; statements about uniform weight refer to the canonical basis.

Example 4.2 (A one-dimensional system). $dF = \varepsilon d \log w F$ has the solution $F = C w^\varepsilon = C(1 + \varepsilon \log w + \frac{\varepsilon^2}{2} \log^2 w + \dots)$: the recursion (18) generates exactly the expansion of the power function. Both descriptions of this object — the expanded logarithmic series and the unexpanded power w^ε — will be needed; the tension between them is the subject of Section 9.

Example 4.3 (A two-dimensional system and the dilogarithm). Take the alphabet $\{w, 1-w\}$ on $0 < w < 1$ and the system

$$d\vec{F} = \varepsilon \left[\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} d \log w + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} d \log(1-w) \right] \vec{F}, \quad (21)$$

with the boundary condition $\vec{F} \rightarrow (1, 0)^\top$ as $w \rightarrow 0$ (the precise sense in which a value is assigned at a singular point is given in Section 4.3). The first component stays constant. For the second, the recursion gives

$$F_2^{(1)}(w) = \int_0^w d \log(1-w') \cdot 1 = \log(1-w), \quad (22)$$

$$F_2^{(2)}(w) = \int_0^w \left[d \log(1-w') \cdot 0 - d \log w' \cdot F_2^{(1)}(w') \right] = - \int_0^w \frac{dw'}{w'} \log(1-w') = \text{Li}_2(w). \quad (23)$$

The dilogarithm appears as the twofold iterated integral with the word $(w, 1-w)$; the next order produces trilogarithms, and so on. Every master of the calculation is, order by order, a rational combination of such words in its own alphabet.

4.3 Local behaviour at the singular loci

The canonical form does more than integrate: it displays, before any integration, how every solution behaves at every singular locus. Near a zero of a letter — say $w \rightarrow 0$, with letter w — the system reads

$$\partial_w \vec{F} = \frac{\varepsilon R_w}{w} \vec{F} + (\text{regular}), \quad (24)$$

and the classical Frobenius analysis applies: for generic ε a fundamental system of local solutions is

$$\vec{F} \sim w^{\varepsilon R_w} \vec{h}(w) = \sum_{\text{modes}} w^{n+a\varepsilon} (\log w)^j (\text{const} + O(w)), \quad (25)$$

with the coefficients a ranging over the eigenvalues of R_w , non-negative integer shifts n from the Taylor expansion of the regular part, and powers of $\log w$ from the Jordan structure of R_w . *The complete local behaviour of every solution at every singular locus is thus encoded in the residue matrix at the corresponding letter* — exponents in its eigenvalues, logarithms in its Jordan blocks — and is known exactly in advance. Matching the global solution onto (25) is also what defines a boundary value at a singular point: the constant in the mode of exponent zero.

Two qualifications keep the statement honest. The normal form requires that no two exponents $\varepsilon\lambda_i, \varepsilon\lambda_j$ differ by a nonzero integer — automatic for generic ε , which is the regime of the Laurent expansion, but not an identity in ε . And “logarithms come from the Jordan blocks” refers to the solution at fixed ε ; upon expanding in ε the logarithms proliferate for a second, independent reason,

$$w^{\varepsilon R_w} = e^{\varepsilon R_w \log w} = \sum_{m \geq 0} \frac{(\varepsilon \log w)^m}{m!} R_w^m, \quad (26)$$

so a power of $\log w$ appears at every order in ε whether or not R_w is diagonalizable; the Jordan structure bounds only the logarithms that survive at fixed ε . This distinction — the unexpanded mode versus its logarithmic series — is precisely what the endpoint analysis of Section 9 turns on.

Remark 4.4 (Existence). *No theorem guarantees that a given block admits a canonical form with rational letters. In the double-real channel every one of the classes admits one — though for some only after a change of variables that rationalizes a square root, and for the classes tied to three independent roots the appropriate generalization is the subject of Section 6. The methodological position throughout is that the canonical form is a statement to be verified: any means whatsoever may produce the transformation, and what establishes it is the exact identity between the transformed connection and the form (16).*

5 Construction of the canonical transformation

The unknown of this section is an invertible matrix $T(v, w; \varepsilon)$ relating the derived basis to the canonical one, $\vec{I} = T \vec{F}$. Substituting into $d\vec{I} = A\vec{I}$ shows that the connection transforms inhomogeneously,

$$A \mapsto A' = T^{-1}AT - T^{-1}dT, \quad (27)$$

and the requirement is that A' be of the form (16). The triangular structure (14) decides the order of the work. A diagonal block never feels what lies below it, so each block can be brought to canonical form on its own — and since the same block recurs across many families, this is done once for each class. With the diagonal canonical, the entries below it yield to transformations of a particularly simple, unipotent shape. What survives both steps is a dependence on ε that is constant in the kinematics, and it is removed last, by conjugation.

5.1 Diagonal blocks: Fuchsian reduction and balances

For the diagonal blocks two complementary strategies are classical, and each produces a candidate transformation which then stands or falls on the exact identity.

The first is the *rational ansatz*: posit for T a matrix of rational functions whose denominators are drawn from the expected letters and whose numerators are polynomials of bounded degree with undetermined ε -dependent coefficients; impose that (27) produce the form (16); solve the resulting linear equations for the coefficients. The approach is exhaustive at a given degree.

The second proceeds through normal forms. One first reduces the connection to *Fuchsian* form — simple poles only — and then normalizes the eigenvalues of the residue matrices by *balance transformations* between a pair of singular points v_1, v_2 ,

$$T = \bar{P} + \frac{v - v_1}{v - v_2} P, \quad \bar{P} = 1 - P, \quad (28)$$

with P the spectral projector of the residue at v_1 onto the eigenspace being moved. Such a transformation lowers the chosen eigenvalue at v_1 by one and raises the matching eigenvalue at v_2 by one, leaving every other singular point untouched; a balance always acts at two points, never one, since the sum of all residue eigenvalues over the singular points (infinity included) is invariant. A sequence of balances aligns all local exponents proportionally to ε , after which a constant transformation factors the regulator out of the connection.

Example 5.1 (What a higher pole means, and what is removable). *Consider first the scalar equation*

$$\frac{dF}{dw} = \left(\frac{\alpha}{w^2} + \frac{\beta \varepsilon}{w} \right) F. \quad (29)$$

The substitution $F = e^{-\alpha/w} G$ does remove the double pole, since $\frac{d}{dw} e^{-\alpha/w} = \frac{\alpha}{w^2} e^{-\alpha/w}$, leaving $G' = (\beta \varepsilon / w) G$ and $G = C w^{\beta \varepsilon}$. But $e^{-\alpha/w}$ is not a rational — not even a meromorphic — change of basis, and no rational one can do the job: for a scalar equation a rational $F = r(w) G$ shifts the connection by $-d \log r$, which has only simple poles with integer residues. A scalar double pole is therefore always a genuinely irregular singularity, with the essential-singular factor $e^{-\alpha/w}$ in its solution, and no canonical form exists.

What is removable is a higher pole sitting in a nilpotent direction of a matrix residue. For

$$A = \frac{1}{w^2} \begin{pmatrix} 0 & \alpha \\ 0 & 0 \end{pmatrix} dw, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & w \end{pmatrix} \implies A' = \frac{1}{w} \begin{pmatrix} 0 & \alpha \\ 0 & -1 \end{pmatrix} dw, \quad (30)$$

a simple pole, reached by a rational transformation — *Fuchsian*, and ready for the balances above. In the present calculation the genuinely irregular case never occurs once the variables of Section 6 are chosen correctly.

5.2 Coupling the blocks: the completion equation

Suppose the diagonal blocks are already canonical — each equal to εe_k with e_k a constant-residue logarithmic form — while the entries below the diagonal, denoted b_{kj} with $j < k$, are not. They are brought to canonical form one block row at a time, by transformations of unipotent shape:

$$T = 1 + D, \quad D \text{ supported on block row } k, \text{ columns } j < k. \quad (31)$$

Because D is strictly triangular, $D^2 = 0$ and $T^{-1} = 1 - D$; moreover DAD and DdD vanish — the latter because D and dD share the same support, the former because A is block lower triangular, so its (j, k) entries with $j < k$ are zero — and the transformation law (27) is exact without expansion:

$$A' = A + AD - DA - dD. \quad (32)$$

Reading off the (k, j) block of the requirement that A' be canonical gives the *completion equation*

$$dD_{kj} = \varepsilon(e_k D_{kj} - D_{kj} c_j) + \bar{b}_{kj} - \varepsilon \sum_a K_a d \log \varphi_a, \quad (33)$$

in which c_j is the canonical form of the diagonal block in column j (so that $a_{jj} = \varepsilon c_j$, in parallel with $a_{kk} = \varepsilon e_k$), the source $\bar{b}_{kj} = b_{kj} - \sum_{j < m < k} D_{km} b_{mj}$ collects the original coupling together with the contributions of blocks of the same row completed previously, and the constant matrices K_a are unknowns alongside D_{kj} : they become the residues of the completed entry. The essential feature of (33) is its *linearity* in the unknowns (D_{kj}, K_a) , which places the whole completion within reach of linear algebra.

Example 5.2 (A completion displaying all three mechanisms). *In one variable, with diagonal data $e = d \log v$ and $c = 0$, take the coupling*

$$b = \varepsilon dv + \frac{dv}{1+v}. \quad (34)$$

The two pieces are absorbed by different mechanisms. For the first, try a rational D : with $D = \frac{\varepsilon v}{1-\varepsilon}$ one has

$$\varepsilon e D + \varepsilon dv = \varepsilon \frac{dv}{v} \cdot \frac{\varepsilon v}{1-\varepsilon} + \varepsilon dv = \frac{\varepsilon}{1-\varepsilon} dv = dD, \quad (35)$$

so the non-logarithmic part of the coupling disappears into the transformation. The second piece is itself a logarithmic form with constant coefficient, $d \log(1+v)$: no rational D can absorb it (that would require $\log(1+v)$ itself), and none should — it is kept as a residue, $K_{1+v} = 1$. It appears at order ε^0 rather than multiplied by ε , but that mismatch is a normalization of the basis, constant in the kinematics, and is removed by the final transformation of Section 5.4. Rational parts of a coupling are absorbed by D ; logarithmic parts with constant coefficients become residues; normalization mismatches in ε are deferred to the end — this division of labour persists in the general case.

5.3 Solution of the completion equation by modular arithmetic

For the systems of interest the unknown D_{kj} may have several hundred rational-function entries of high degree, and solving (33) symbolically is prohibitive. The equation is linear, however, and a linear problem can be solved numerically over finite fields and reassembled exactly. One writes $D_{kj} = N(v, w; \varepsilon) / \prod_a \varphi_a^{m_a}$, with the denominator a product of letters and the numerator a polynomial with undetermined coefficients, the admissible monomials being bounded by a degree analysis of the source $\bar{b}_{kj}$. Substituting rational values of (v, w) and of the regulator, all reduced modulo a large prime p , turns (33) into exact linear equations over the finite field $\mathbb{F}_p$ for the unknown coefficients; enough substitutions determine them modulo p , and the dependence on ε — the solution is a rational function of the regulator as well — is recovered by sampling sufficiently many regulator values and interpolating. Repeating over several primes, combining by the Chinese remainder theorem, and lifting each coefficient to a rational number by rational reconstruction yields an exact candidate, confirmed on evaluations withheld from the fit and at a fresh prime. None of this is a proof; the proof is the substitution of the reconstructed (D, K) back into (33) — or into the equivalent statement for the fully assembled family — and the exact verification of the identity.

Example 5.3 (Rational reconstruction). *Take $p = 101$ and suppose a coefficient is determined as $29 \bmod 101$. Rational reconstruction seeks a fraction of small numerator and denominator congruent to it: since $7 \cdot 29 = 203 = 2 \cdot 101 + 1 \equiv 1 \pmod{101}$, one has $29 \equiv \frac{1}{7} \pmod{101}$. A second prime, $p = 103$, where $\frac{1}{7} \equiv 59 \pmod{103}$ (as $7 \cdot 59 = 413 = 4 \cdot 103 + 1$), and the Chinese remainder lift of $(29, 59)$ modulo $101 \cdot 103$ reconstructs $\frac{1}{7}$ again. The guarantee behind the search is a lattice*

bound: modulo a single prime p the reconstruction is unique only for numerator and denominator below $\sqrt{p}/2$ — about 7.1 for $p = 101$, so $\frac{1}{7}$ is admissible there only just, which is exactly why the second prime is not optional. Stability of the lift as the modulus grows, plus agreement at a prime never used, is compelling evidence; the proof remains the final exact substitution.

Remark 5.4 (Non-uniqueness of the completion). *When the linear system for (D, K) has a null space the completed entry is not unique; all solutions are related by redefinitions that preserve the canonical form. All are correct, but they differ substantially in the degrees of the rational functions they insert into the sources $\bar{b}$ of the rows still to be completed, and degree growth compounds from row to row. A definite and favourable choice results from requiring the coefficients of the highest admissible numerator monomials to vanish, which selects a representative of low degree. The choice leaves no trace in the final canonical form; it decides only the degrees that the remaining rows inherit.*

5.4 The residual dependence on the regulator

After the completion, the connection is logarithmic with ε -independent letters, but its residue matrices may still depend on ε — rational functions of ε in place of the required constants times ε . The origin of this residue is the normalization of the master basis: each sector's masters were defined with ε -dependent prefactors, and a change of normalization is a transformation *constant in the kinematics*. For constant $T(\varepsilon)$ the inhomogeneous term of (27) vanishes and the transformation acts by simultaneous conjugation of all residues:

$$T^{-1}(\varepsilon) \tilde{R}_a(\varepsilon) T(\varepsilon) = \varepsilon R_a \quad \text{for all } a. \quad (36)$$

Finding such a T is a finite-dimensional problem: evaluate the connection at a few exact rational kinematic points, factor the ε -dependence of the resulting constant matrices, and verify the candidate by the exact identity on the full connection. Composing the class transformations, the row completions, and this constant conjugation yields the matrix T that carries the derived system (7) into the canonical form (16).

6 Algebraic letters: square roots and their rationalization

6.1 Square roots from the geometry of phase space

Nothing so far required the letters to be polynomials, and for a minority of the blocks they are not. The origin is geometric. Certain sub-configurations of the three-particle final state degenerate on the vanishing of a Gram determinant, which in the variables (v, w) is the Källén polynomial

$$\lambda_1(v, w) = (1 - v - w)^2 - 4vw = \lambda(1, v, w), \quad (37)$$

where $\lambda(a, b, c) = a^2 + b^2 + c^2 - 2ab - 2bc - 2ca$ is the triangle function. Its zero locus is the curve $\sqrt{v} + \sqrt{w} = 1$, a pseudo-threshold lying *inside* the physical triangle (3) rather than on its boundary — the point $v = w = \frac{1}{4}$ sits on it — and it separates the region $\lambda_1 > 0$, where $\sqrt{\lambda_1}$ is real, from the rest. Masters of sectors sensitive to this locus have branch points of square-root type there, and a square-root branch point cannot be produced by logarithmic differentials of polynomials alone: the alphabet must contain $\sqrt{\lambda_1}$, typically through letters of the form

$$\frac{1 - v - w - \sqrt{\lambda_1}}{1 - v - w + \sqrt{\lambda_1}}, \quad (38)$$

which are exchanged with their reciprocals under the sign change $\sqrt{\lambda_1} \rightarrow -\sqrt{\lambda_1}$ (in the language of covering spaces, the deck transformation of the double cover). The double-real channel realizes six such radicands:

$$\lambda_1, \quad \lambda_2 = \lambda_1(-v, w), \quad \lambda_3 = \lambda_1(v, -w), \quad 4v + w^2, \quad v^2 + 4w, \quad 1 - 4vw. \quad (39)$$

Under the involution $v \leftrightarrow w$, the pairs $\{\lambda_2, \lambda_3\}$ and $\{4v + w^2, v^2 + 4w\}$ are exchanged, while λ_1 and $1 - 4vw$ are invariant.

Definition 6.1 (Root content of a system). *Radicands are counted modulo perfect-square factors and modulo products: $\sqrt{q_1}$, $\sqrt{q_2}$ and $\sqrt{q_1 q_2}$ constitute two independent roots, since the third is determined by the other two. Formally, the radicands of a system generate a subgroup of the square-class group $\mathbb{Q}(v, w)^\times / (\mathbb{Q}(v, w)^\times)^2$, and the root content is the rank of that subgroup. In the double-real channel most families involve no root at all; a third involve one; a dozen involve two; and three families — the hard case treated in Section 6.4 — involve three.*

The root content is read off the derived connection and reduced exactly; it is the single datum that determines which of the techniques below applies.

6.2 Rationalization by change of variables

Definition 6.2 (Rationalizing substitution). *A rationalizing substitution for a radicand $q(v, w)$ is a rational change of variables $(v, w) = (f(x, y), g(x, y))$ with non-degenerate Jacobian such that $q(f, g) = r(x, y)^2$ for a rational function r . Under such a substitution $\sqrt{q}$ becomes the rational function r , and a system whose alphabet involves only $\sqrt{q}$ acquires polynomial letters in (x, y) .*

Example 6.3 (Rationalizing the Källén root). *Consider the substitution suggested by the factorized form of the degenerate configurations,*

$$v = xy, \quad w = (1 - x)(1 - y). \quad (40)$$

Then

$$1 - v - w = x + y - 2xy, \quad \lambda_1 = (x + y - 2xy)^2 - 4xy(1 - x)(1 - y) = (x - y)^2, \quad (41)$$

by direct expansion. Hence $\sqrt{\lambda_1} = x - y$, the Jacobian determinant is $\partial(v, w)/\partial(x, y) = x - y$, non-vanishing away from the diagonal, and the algebraic letter above becomes

$$\frac{(x + y - 2xy) - (x - y)}{(x + y - 2xy) + (x - y)} = \frac{y(1 - x)}{x(1 - y)}, \quad (42)$$

a ratio of polynomial letters. Two facts about the substitution deserve recording. It is two-to-one: both v and w are symmetric under $x \leftrightarrow y$, and that exchange is precisely the deck transformation $\sqrt{\lambda_1} \rightarrow -\sqrt{\lambda_1}$, branched on the diagonal $x = y$ where the Jacobian vanishes. And its real image is not the whole triangle (3) but the sub-region $\sqrt{v} + \sqrt{w} \leq 1$, i.e. $\lambda_1 \geq 0$; outside it $\sqrt{\lambda_1}$ is imaginary, and reaching it is an analytic continuation to be performed deliberately. Paths chosen in (x, y) therefore live in that sub-region. The sign variants $v = \pm xy$, $w = \mp(1 - x)(1 - y)$ rationalize λ_2 and λ_3 in the same way, and substitutions of the same conic type rationalize $4v + w^2$, $v^2 + 4w$ and $1 - 4vw$ — one substitution for each of the six radicands.

The construction behind all such substitutions is classical: the equation $q(v, w) = r^2$ defines a conic in suitable projective coordinates, and a conic possessing one rational point is rationally parametrized by the pencil of lines through that point — the same device that parametrizes the Pythagorean triples. For each radicand of (39) the substitution can be written down explicitly, together with its root image and Jacobian, and the identity $q \circ (f, g) = r^2$ is then a matter of expansion, never taken on faith.

Working in the new variables requires one transformation of the whole system by the chain rule,

$$A_x = A_v \partial_x f + A_w \partial_x g, \quad A_y = A_v \partial_y f + A_w \partial_y g, \quad (43)$$

with flatness re-established in (x, y) ; afterwards the constructions of Sections 4–5 apply verbatim.

6.3 Several roots

A system whose alphabet involves two independent roots requires a substitution rationalizing both at once. Such substitutions are found iteratively: rationalize the first root as above; under the substitution, the second radicand becomes a new polynomial $q_2(x, y)$, defining a second conic *in the new variables*; a rational point on that conic yields a parametrization by lines, in one of the variables, that rationalizes it while preserving the rationality of the first root. Three iterated substitutions of this kind — one for each pair of Källén-type radicands — suffice for every two-root family that occurs, and the resulting canonical connections are naturally written in the final pair of parameters.

6.4 The Galois algebra of the roots

For the families involving three independent roots, no rational substitution can rationalize all three simultaneously; this is a theorem, not a failure of search. A rationalizing substitution is a dominant rational map from the plane, so its existence would make the fibred product of the three conic double covers a unirational surface; for the root triples occurring here that surface has two ordinary double points whose resolution is a K3 surface, which is not unirational in characteristic zero. The obstruction is geometric and definitive.

The response is to renounce parametrization and compute in the extension the roots generate. Adjoin them as symbols:

$$E = \mathbb{Q}(v, w, \varepsilon) [r_1, r_2, r_3] / (r_i^2 - q_i), \quad (44)$$

an algebra of dimension 2^3 with basis $r_S = \prod_{i \in S} r_i$ indexed by subsets $S \subseteq \{1, 2, 3\}$; when the q_i are independent square classes — which is what root content three means — E is a field extension of degree eight. Its Galois group is $(\mathbb{Z}/2)^3$, acting by the independent sign changes $r_i \mapsto -r_i$. Every element decomposes into the characters of this group (its *grades*); multiplication respects the grading,

$$r_S r_T = \left(\prod_{i \in S \cap T} q_i \right) r_{S \Delta T}, \quad (45)$$

and differentiation acts grade-wise through $dr_S = \frac{1}{2} r_S \sum_{i \in S} d \log q_i$. All linear problems of Section 5 — the completion equation, its evaluation over finite fields, the reconstruction — can be posed on the 8-component coefficient vectors, whose entries are ordinary rational functions. Evaluations modulo a prime are performed at points where every q_i is a nonzero quadratic residue, so that all eight embeddings of the algebra into $\mathbb{F}_p$ exist and the grade decomposition becomes a

discrete Fourier transform over $(\mathbb{Z}/2)^3$; agreement of a solution across all eight sign embeddings is a consistency requirement that a correct computation must, and does, satisfy. No better description of these systems is presently known; the construction has been carried through on model systems of every root rank, and the physical three-root families are of the same type.

Remark 6.4 (Roots and the crossing involution). *The involution $v \leftrightarrow w$ that enters the equivalence of blocks (Definition 3.3) exchanges λ_2 with λ_3 and $4v + w^2$ with $v^2 + 4w$ while fixing λ_1 and $1 - 4vw$; within the Källén substitution it takes the simple form $(x, y) \mapsto (1 - x, 1 - y)$.*

7 The general solution: paths and polylogarithms

The canonical form reduces integration to the evaluation of the iterated integrals in (20) along paths in the physical region. For letters *linear* in the integration parameter the iterated integrals belong to a standard class:

Definition 7.1 (Generalized polylogarithms). *For $a_1, \dots, a_k \in \mathbb{C}$ define recursively*

$$G(a_1, a_2, \dots, a_k; t) = \int_0^t \frac{dt'}{t' - a_1} G(a_2, \dots, a_k; t'), \quad G(; t) = 1, \quad (46)$$

with the logarithmic case $G(\vec{0}_k; t) = \frac{1}{k!} \log^k t$. These generalized polylogarithms of weight k contain the classical functions: $G(0, 1; t) = -\text{Li}_2(t)$, matching Example 4.3 up to the sign conventions, and all functions of weight two and three reduce to classical polylogarithms.

Evaluating a family is then a matter of choosing a base point p_0 in the triangle (3) and a path to the point of interest, composed of segments on which every letter restricts to a polynomial of degree one in the path parameter — for then the solution along each segment is a combination of the functions G . When a letter restricts to a quadratic, subdividing the segment does not help: the restriction stays quadratic on every piece. The remedy is either to choose the segment differently — in practice segments parallel to the coordinate axes, on which the bilinear letters produced by the rationalizing substitutions become linear again — or to factor the quadratic and admit its two linear factors, with algebraic constants, as letters in their own right. Flatness guarantees that the result depends only on the endpoint and on the homotopy class of the path; the zero locus of a letter is crossed only as a deliberate analytic continuation, never in passing. Families whose canonical form lives in rationalizing variables are evaluated in those variables throughout, along paths chosen by the same linearity criterion, and only the final values are mapped back to (v, w) .

7.1 The order of expansion required

The cost of an iterated integral grows steeply with its weight, so one does not expand every master to the same depth; the required depth is dictated by where the master enters the cross section. A master multiplying a double pole of a coefficient needs two orders in ε beyond one multiplying a finite coefficient, and the required order is found for each master by following its coefficients through the assembled combination. In the double-real channel the outcome is lopsided: of the 347 masters, 203 are needed only to their finite order and 131 one order deeper, while 9 are needed deeper still and the remaining four less deeply (two enter only at order ε , and two multiply coefficients that vanish identically). This accounting is settled before any integration, and it sets the scale of the entire computation.

8 Boundary conditions

8.1 The space of solutions and the conditions upon it

The path-ordered solution (20) involves the undetermined vector $\vec{F}(p_0) = \sum_k \varepsilon^k \vec{C}^{(k)}$: one constant per master per order. Determining these constants is where analysis meets physics, and the governing principle is that almost all of them are fixed by consistency, without evaluating a single new integral.

The most productive condition — derived in the next subsection — is the *exclusion of local behaviour*. Near each edge of the physical triangle the differential equation offers local modes with exponents read from its residue matrices (Section 4.3); the integral representation of the master permits only some of them; every offered mode that the representation forbids annihilates one combination of constants.

Three further conditions complete the analysis. Elementary normalizations: the simplest master of the lowest sector is the phase-space volume itself, computable in closed form (Example 8.2), and its expansion provides initial data wherever that sector appears. Inheritance: families are analysed in an order compatible with the subsector relation, and a master of a subsector, once known, is known in every family in which that subsector reappears — the identification of Section 3 is an equality of integrals, not an analogy — imposing linear conditions there. Finiteness at spurious loci: some zero loci of letters lie inside the region where a given integral is manifestly finite, and the local modes singular there must then be absent from it.

8.2 Power counting at the boundary

Consider a master given by a convergent cut phase-space integral, and examine one edge of the triangle, say $w \rightarrow 0$. In the limit the integral is governed by a small set of momentum regions — hard, soft, collinear to one or the other direction — and each region contributes to the integral with a definite overall scaling in w . Consequently the exact function has an expansion in the modes

$$w^{n+a\varepsilon}, \quad n \in \mathbb{Z}_{\geq n_0}, \quad (47)$$

where the admissible values of a form a short list fixed by the region analysis: which lines of the sector can become soft or collinear in the limit, and with what measure. It is a coarse datum — a handful of integers and multiples of ε , common to whole groups of sectors — though it does depend on the propagator content of the sector, since that is what decides which regions exist. (Over the whole double-real channel, for instance, the spectrum of ε -coefficients on the soft edge consists of just five values.) The differential equation, on the other hand, offers local modes with exponents drawn from the eigenvalues of the residue matrix at the corresponding letter. Whenever the equation offers an exponent that the power counting excludes, the coefficient of the entire corresponding mode must vanish: one exact, homogeneous, linear condition on the boundary constants per excluded mode. The same analysis applies at $v \rightarrow 0$ and on the soft edge $1 - v - w \rightarrow 0$, with the distance to the edge as the scaling variable. Two cautions are essential in practice. The admissible exponents must be derived in the same variables in which the local analysis is performed — a rationalizing substitution alters the measure by its Jacobian and shifts the integer parts — and degenerate (Jordan) directions of the residue matrix couple modes logarithmically, so the exclusion argument must be applied to generalized eigenspaces, not to eigenvalues alone.

8.3 The kernel of the constraint system

All conditions on a given block assemble into a linear system on its boundary constants; call its matrix C . The number of genuinely undetermined constants is $\dim \ker C$, and it is known before a single integral is evaluated: it is a property of the constraint matrix, not something that emerges as the calculation proceeds. Constants surviving in the kernel are then identified across blocks — the same subsector appearing in several families contributes one constant, not several — which reduces the count over the whole problem to a short list of genuinely new numbers.

Definition 8.1 (Boundary periods). *The surviving constants are values of cut phase-space integrals at distinguished limits of the kinematics, taken in a definite order: boundary periods. At the weights relevant here they evaluate to rational combinations of ζ -values and kindred constants. Each is established by an exact derivation — through a Mellin–Barnes representation (trading the integrand for a contour integral of Γ -functions), an expansion by momentum regions, or hypergeometric summation — with high-precision numerical evaluation used to corroborate, never to prove.*

Example 8.2 (The phase-space volume). *The volume of the massless n -particle phase space in d dimensions is fixed by Lorentz invariance and dimensional analysis up to a constant, and iterating the two-body formula gives it in closed form:*

$$\int \prod_{i=1}^n d^d p_i \delta^{(+)}(p_i^2) \delta^d\left(q - \sum_i p_i\right) = (q^2)^{(n-1)(d/2-1)-1} \left(\frac{\pi^{d/2-1}}{2}\right)^{n-1} \frac{\Gamma(d/2-1)^n}{\Gamma((n-1)(d/2-1)) \Gamma(n(d/2-1))}, \quad (48)$$

a pure ratio of Γ -functions; other normalizations of the measure differ by powers of 2π only. (At $d=4$ and $n=3$ this reduces to the familiar $\Phi_3 = \pi^2 q^2/8$ in the convention above.) Three features make it the natural starting point of the boundary analysis: it is the unit-power master of the lowest sector; it requires no differential equation; and its ε -expansion supplies exact ζ -values that propagate upward through the inheritance conditions. The genuinely new periods appearing higher in the partial order are of the same character — cut integrals at an ordered limit, reducible to Γ -functions or to low-dimensional Mellin–Barnes integrals.

Example 8.3 (The two-dimensional system, concluded). *For the system of Example 4.3 the boundary vector has two components per order. At $w \rightarrow 0$ the residue matrix has eigenvalues $\{0, -1\}$, offering local exponents $\{0, -\varepsilon\}$; if the underlying integral is finite and vanishing at $w=0$, the mode $w^{-\varepsilon}$ is excluded and one constant is annihilated. The other is inherited: the first component is the volume-type subsector master, already known. The kernel of the constraint system is trivial, no new period appears, and the solution $\text{Li}_2(w)$ carries no free constant — in miniature, exactly the accounting of the full calculation.*

9 Behaviour at the edges of phase space

The local analysis of Section 4.3 identified, within the global solution, the coefficient of each singular mode at each edge. One further step connects these modes to the cross section. The cross section integrates the masters against smooth functions up to the boundary, and near an endpoint the relevant integrals contain modes of the form $(1-w)^{-1+a\varepsilon}$. For such a mode the expansion in ε and the integration against a test function do not commute. The correct object is

the distributional identity

$$(1-w)^{-1+a\varepsilon} = \frac{1}{a\varepsilon} \delta(1-w) + \sum_{k \geq 0} \frac{(a\varepsilon)^k}{k!} \left[\frac{\log^k(1-w)}{1-w} \right]_+, \quad (49)$$

where the *plus distribution* is defined by

$$\int_0^1 dw [g(w)]_+ f(w) = \int_0^1 dw g(w) [f(w) - f(1)] \quad (50)$$

for test functions f smooth at $w = 1$. The proof of (49) is one line: integrate against f , add and subtract $f(1)$,

$$\int_0^1 dw (1-w)^{-1+a\varepsilon} f(w) = \frac{f(1)}{a\varepsilon} + \int_0^1 dw (1-w)^{-1+a\varepsilon} [f(w) - f(1)], \quad (51)$$

where the first term uses $\int_0^1 (1-w)^{-1+a\varepsilon} dw = 1/(a\varepsilon)$, convergent for $\text{Re}(a\varepsilon) > 0$ and continued from there; the second integrand may be expanded in ε term by term because $f(w) - f(1)$ vanishes at $w = 1$, making every $\log^k(1-w)/(1-w) \cdot [f(w) - f(1)]$ integrable. Modes $(1-w)^{n+a\varepsilon}$ with $n \leq -2$ are treated by the same subtraction carried to higher order and produce derivatives of $\delta(1-w)$.

Had one expanded the *mode* first, the left-hand side would have become the series $\frac{1}{1-w} \sum_k \frac{(a\varepsilon)^k}{k!} \log^k(1-w)$, each term of which is non-integrable at $w = 1$: the δ -distribution — the very content the cross section needs — is destroyed by premature expansion. For this reason the analytic evaluation carries, next to the expanded polylogarithmic form of each master, its exact local data: which exponents $n + a\varepsilon$ occur at the endpoint loci and with which coefficients. The identity (49) is applied to the unexpanded modes, and only afterwards is everything expanded in the regulator. Recovering the endpoint behaviour is thus not an act of resummation but of refraining: the residue matrices supplied the exponents, the boundary analysis supplied the coefficients, and (49) converts them into distributions.

10 Search and proof

Nothing in the construction depends on how a transformation was found. An ansatz, a numerical fit, arithmetic modulo a prime — each produces a candidate and nothing more; what makes the candidate a theorem is the identity $T^{-1}AT - T^{-1}dT = \varepsilon \sum_a R_a d \log \varphi_a$, verified exactly. The same asymmetry runs through the whole subject: flatness establishes a connection, exact reconstruction from constant residues establishes a canonical form, and a boundary period enters only through a derivation in which numerics appear afterwards, as a check on arithmetic, never as evidence. Where a verification is performed modulo primes at random points, its failure probability is bounded and stated, and it is the stated bound — not an impression of reliability — that the result carries.

It is honest to record the state of the calculation these notes describe. At the time of writing, all but the three three-root families carry canonical forms established in this way — eighty-eight of the ninety-one — while the three-root families await the Galois-algebraic treatment of Section 6.4; of the boundary periods, the lowest tier is established and the remainder is an open campaign, as is the endpoint mode analysis of the full inventory. The method is complete; the calculation, like every large calculation, is a living object.

11 Notation and terminology

Integral family / representative family

A complete set of inverse propagators (5) / its representative modulo cut-preserving redefinitions of the loop momenta.

Sector, subsector

The subset of propagators present with positive power; a subsector arises by deleting lines.

Master integral

Basis element of a family's integrals modulo integration-by-parts identities.

Connection, flatness

The matrix one-form A in $d\vec{I} = A\vec{I}$; the integrability condition $dA = A \wedge A$.

Block, class

A strongly connected group of masters (a diagonal block of the triangular connection); its equivalence class under basis permutation composed optionally with $v \leftrightarrow w$.

Canonical form (ε -form), letters, alphabet, residues

The form $A = \varepsilon \sum_a R_a d \log \varphi_a$ with constant R_a ; the functions φ_a ; their set; the matrices R_a .

Completion equation

The linear equation (33) for the transformation of the entries below the diagonal.

Root content, rationalizing substitution

The rank of the group of square classes in an alphabet; a rational change of variables under which roots become rational.

Galois algebra of the roots

The extension $\mathbb{Q}(v, w, \varepsilon)[r_1, \dots, r_k]/(r_i^2 - q_i)$ with automorphism group $(\mathbb{Z}/2)^k$ acting by sign changes.

Boundary period

A genuinely new constant surviving all boundary conditions; the value of a cut phase-space integral at an ordered limit.

Local modes

The exponents $n + a\varepsilon$ and logarithm powers of the Frobenius solutions at a singular locus, read from the residue matrix.

Further reading. Henn's paper introducing the canonical form is short and repays being read first; Duhr's lectures are the standard entry to iterated integrals and polylogarithms, including the shuffle and stuffle structure these notes pass over; the differential-equation method itself goes back to Kotikov and to Gehrmann and Remiddi, and its algorithmic reduction to canonical form to Lee (with Meyer's and Lee's implementations the practical references); Anastasiou and Melnikov introduced reverse unitarity for exactly the present kind of phase-space problem; Besier, Wasser and Weinzierl treat the rationalization of square roots systematically; Pak's algorithm is the standard reference for the normal ordering of Symaznik polynomials.